

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE ELECTRICAL,
MECHANICAL, CIVIL, PETROLEUM, AEROSPACE, BIOMEIDCAL & BIOSYSTEMS
ENGINEERING

ECU 201: ENGINEERING MATHEMATICS IV1ST SEMESTER 2020/2021

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO Questions.

QUESTION ONE:

- (a) The following table gives the frequency distribution of ages of all 50 employees of a company.

Age	Number of employees
18 – 30	13
31 – 43	17
44 – 56	14
57 – 69	6

- (i) Determine the class width of the third age category. (1 mark)
- (ii) Compute the relative frequencies. (2 marks)
- (iii) Find the percentage of employees that are older than 43 years. (2 marks)
- (b) The following give the number of hours spent doing homework and attending classes by 10 randomly selected college students during the past one week.
- 8 6 17 24 29 25 32 19 21 24
- Calculate median, mode and variance for these data. (5 marks)
- (c) The probability that a family owns a desktop computer is 0.78, that it owns a laptop is 0.7 and that it owns both the computer and a laptop is 0.6. Find the probability that a randomly selected family owns a computer or a laptop. (3 marks)
- (d) Fifty students took up a test. The results of those who passed the test is given below:

Marks	4	5	6	7	8	9
Number of students	8	10	9	6	4	3

If the average mark of all fifty students was 5.16, find the average mark of those who failed.

(4 marks)

(e) A quality test engineer claims that 1 in 10 of certain manufactured parts is due to material defects. Using binomial distribution and rounding to four decimals, find the probability that at least 2 out of 4 of the tested parts are due to material defects. (4marks)

(f) The number of defects on an electronic component which is used in a computerized system has been found to follow the Poisson distribution with $\lambda=3$. Find the probability that a randomly selected electronic component will have at most two defects. (4marks)

(g) Customers buying computer parts supplied by a certain manufacturer require that the parts be between 9.9cm and 10.5 cm, inclusive. The manufacturing process is such that the actual part lengths are well approximated by a normal distribution with mean 10.1 cm and standard deviation 0.20 cm. Determine the percentage of the manufacturer's production that is acceptable to the customer. (5marks)

QUESTION TWO

(a) The following table gives the number of telephone calls received per hour in an office and the probability function.

No. of telephone calls Received, x	10	11	12	13	14	15
f(x)	0.08	0.15	0.30	0.20	0.20	0.07

(i) Find the probability that there will be more than 13 calls per hour. (3marks)

(ii) Determine the mean and standard deviation of the number of telephone calls received per hour in that office. (7marks)

(b) The following data give the number of machines manufactured by a company for a sample of 30 days:

28 21 31 32 26 35 27 31 34 23 22
 21 26 27 26 33 33 31 29 27 23 25
 33 35 23 28 31 28 29 26

(i). Construct a frequency distribution table using classes 21-23, 24-26, 27-29, 30-32 and 33-35. (2marks)

Use the frequency distribution to calculate

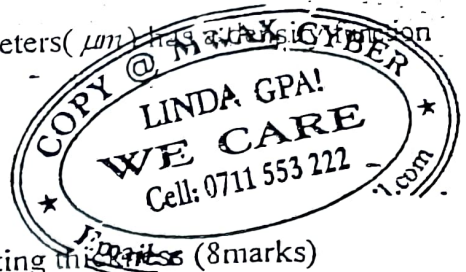
(ii). Mean, mode and median of the distribution. (8marks)

1st SEM 2020/2021

QUESTION THREE

(a) The thickness of a conductive coating in micrometers (μm) has a probability density function

$$f(x) = \begin{cases} \frac{600}{x^2}, & 100 \mu m < x < 120 \mu m \\ 0, & \text{elsewhere} \end{cases}$$



- (i) Find the mean and variance of the coating thickness (8marks)
- (ii) If the cost of a coating is ksh.25 per micrometer of thickness on each component, find the average cost of the coating per component (2marks)

(b) The weight of an electric component is normally distributed with a mean of 6.8 mm and a standard deviation of 0.25 mm

- (i) Find the probability that the electronic weighs more than 7mm. (3marks)
- (ii) What must the standard deviation of weight be in order for the company to state that 90% of its electronic components are less than 7mm. (4marks)
- (iii) If the standard deviation stays at 0.25 mm what must the mean weight be in order for the company to state that 99% of its electronic components are less than 7mm. (4marks)

QUESTION FOUR

(a) The following data represent the amount of corrosion coating (in ounces) needed to paint six plates:

8.2 8.8 7.7 7.9 8.4 and 7.9

Determine the following:

- (i) The point estimate of μ , the mean amount of corrosion coating needed to paint a plate and also the point estimate of the variance. (4 marks)
- (ii) 98% confidence for the mean assuming that the amount of corrosion coating per plate is normally distributed and that standard deviation of all the corrosion coating is 0.38 ounces. (6marks)
- (b) A random variable X has a Bernoulli distribution whose density function is given by

$$f(x) = \begin{cases} p^x (1-p)^{1-x}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the moment generating function of X and hence find mean and variance of X.

(10marks)

QUESTION FIVE

A chemical company, wishing to study the effect of extraction time, in minutes, on the efficiency of an extraction operation, in %, obtained the data shown in the following table.

Extraction time(x)	27	45	41	19	35	39	19	49	15	31
Extraction efficiency(y)	57	64	80	46	62	72	52	77	57	68

- (a) (i) Construct a scatter diagram for these data
(ii) What relationship does the scatter plot suggest (4marks)
- (b) Find the Pearson's correlation coefficient between extraction time and extraction efficiency. (6marks)
- (c) Determine the least squares regression of extraction efficiency on extraction time (6marks)
- (d) Predict the value of the extraction efficiency at specific times 20 and 30 minutes. (4marks)

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN CIVIL ENGINEERING

ECU 201: ENGINEERING MATHEMATICS VI

DATE: TUESDAY 6TH DECEMBER 2016

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS: Answer Question ONE and any other two Questions.

Q1. (a) The following data gives the number of vehicles owned by each of the 30 families in

a city: 2, 3, 1, 1, 1, 2, 4, 3, 1, 1,
3, 2, 2, 1, 2, 3, 2, 3, 1, 3,
4, 1, 3, 4, 1, 1, 1, 3, 2, 2

(i) Prepare a frequency distribution table for this data set using single-valued classes

(ii) Calculate the mean number of vehicles per family

(iii) Find the percentage of the families in this sample having two or three vehicles.

(3,3,2 marks)

(b) The following are the ages of 15 employees of a manufacturing company:

65, 38, 52, 27, 24, 45, 49, 50, 26, 37, 44, 52, 41, 60, 38

Calculate the mode, median and standard deviation (5marks)

(c) In a group of 10 adults, 4 have type A personality and 6 have type B personality. If two persons are selected at random from this group of 10, what is the probability that the first of them has type A personality and the second has type B personality? Draw a tree diagram for this experiment. (4marks)

(d) The probability that a family owns a desktop computer is 0.78, that it owns a CD player is 0.70 and that it owns both the computer and a CD is 0.60. Find the probability that a randomly selected family owns a computer or a CD player. (3marks)

(e) A student's campus newspaper in a certain university contains an average of 1.2 typographical errors per page. Find the probability that a randomly selected page of this newspaper will contain exactly 4 typographical errors using the Poisson distribution.

(4marks)

(f) The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 7000kg/cm² and a standard deviation of 100kg/cm².

(i) find the probability of a sample's strength that is less than 7250kg/cm²

(ii) find the probability that a sample's strength is between 6800 and 6900kg/cm².

(6mks)

QUESTION TWO

(a) The following data gives the number of car oil changes made at a service station for a sample of 30 days.

23 31 26 24 ~~34~~ ~~34~~ 30 26 24 27
 21 27 30 23 28 32 27 23 27 30
~~33~~ 23 27 27 24 ~~35~~ 30 28 ~~35~~ 29

- Construct a frequency distribution table using the classes $21 - 23$, $24 - 26$, $27 - 29$, $30 - 32$ and $33 - 35$ (4marks)
- Compute the mode and median of the distribution. (6marks)
- Find the average number of oil changes and the standard deviation. (7marks)
- Draw the histogram corresponding to the distribution above together with the frequency polygon. (3marks)

QUESTION THREE

(a) The following table shows the probability distribution of the number of breakdowns per week for a machine.

Breakdowns per week	0	1	2	3
Probability	0.15	0.20	0.35	0.3

- Find the mean and standard deviation of the number of breakdowns
 - Find the probability that the number of breakdowns for this machine during a given week is
 - Exactly 2
 - More than 1
 - At most 1
 (10marks)
- (b) The thickness of a conductive coating in micrometers has a density function(pdf)

$$f(x) = \begin{cases} \frac{600}{x^2}, & 100\mu m < x < 120\mu m \\ 0, & \text{elsewhere} \end{cases}$$

- Find the mean and variance of the coating thickness
- If the cost of coating is \$0.25 per micrometer of thickness on each component, find the average cost of coating per component.
- Find the probability that the coating is more than 115 micrometers.

(10marks)

QUESTION FOUR

- (a) Suppose the following data shows the number of the problems from the Practice Problems Set attempted by 10 randomly selected students:

2, 4, 0, 7, 1, 2, 0, 3, 2, 1

Estimate the mean number and variance of practice problems attempted by a student in the past week. (4marks)

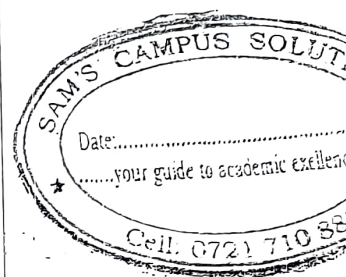
- (b) According to a particular survey, the probability is 0.40 that a traffic fatality involves an intoxicated or alcohol-impaired driver or non-occupant. In 8 traffic fatalities, and using a binomial distribution, find the probability that the number, X , which involve an intoxicated or alcohol-impaired driver or is

- (i) Exactly 3
- (ii) At least 1.
- (iii) Find and interpret the mean of the random variable X .
- (iv) Obtain the standard deviation. (8marks)

- (c) If X has a Bernoulli density function given by

$$P(X = x) = p_x = \begin{cases} p^x (1-p)^{1-x}, & x = 0, 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find the moment generating function of X . Hence find the mean and variance of X . (8marks)



QUESTION FIVE

- (a) A car manufacturing company wanted to investigate how the price of one of its car models depreciates with age. The company took a sample of eight cars of this model and collected the following information on the ages (in years) and prices (in hundreds of dollars) of these cars as shown below:

Age (x)	8	3	6	9	2	5	6	3
Price (y)	16	74	40	19	120	36	33	86

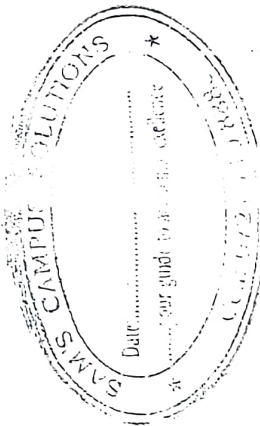
- (i) Construct a scatter diagram for these data and interpret the relationship between age and price of the car. (4marks)
 - (ii) Find the Pearson's correlation coefficient between age and price (4marks)
 - (iii) Find the regression line that would estimate the price given age and use it to estimate the price of a 7 year old car. (6marks)
- (b) The probability that a college student being male is $1/3$ and that of being female is $2/3$. The probability that a male student completes the probability and statistics course successfully is 0.8 and that the female student does it is 0.9. A student is selected at random and is found to have completed the course. Determine the probability that student is a female. (6marks)

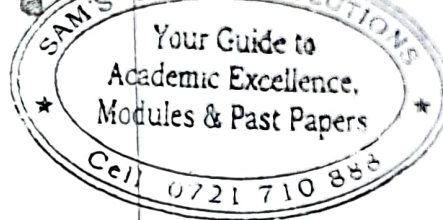
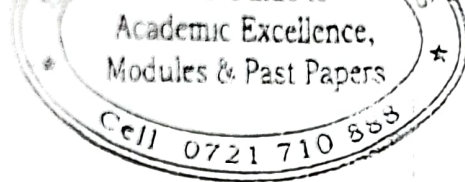
Table 2 CUMULATIVE NORMAL DISTRIBUTION

$$\Phi(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

x	1.282	1.645	1.960	2.326	2.576	3.090	3.291	3.891	4.417
$\Phi(x)$	.90	.95	.975	.99	.995	.999	.9995	.99995	.999995
$2[1 - \Phi(x)]$	.20	.10	.05	.02	.01	.002	.001	.0001	.00001





KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2015/2016

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE

ECU 201: ENGINEERING MATHEMATICS VI

DATE: Wednesday, 16th December 2015

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS:

ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

1. a) The numbers 3.2, 5.8, 7.9 and 4.5 have frequencies x , $x+2$, $x-3$, and $x+6$ respectively. If the arithmetic mean is 4.876, find the value of x . (4 Marks)
- b) The average grade of all the students in class 3X is 68.4 and of all the students in class 3Y is 71.2. If the average of both classes is 70, what is the ratio of students of class 3X to the number in class 3Y (3 Marks)
- c) The first two moments of a distribution about the value 2 are 1 and 16. Calculate the mean and variance of the distribution. (4 Marks)
- d) A carton contains five blue balls and three red balls. Three balls are drawn without replacement. What is the probability of drawing at least one red ball. (4 Marks)
- e) Two events A and B are independent. If $P(A) = \frac{3}{10}$, $P(A \cup B) = \frac{1}{2}$
find (i) $P(B)$
(ii) $P(A \cap B)$ (4 Marks)

f) Weights of objects were found to be normally distributed with mean 45 and standard deviation 14. Calculate the probability that the weights will be between 30 and 70. (4 Marks)

g) The average number of misprints per page of a book is 1.5. If the number misprints follow a Poisson distribution, obtain the probability that at most one misprint on a selected page will occur. (4 Marks)

h) The probability that a person who undergoes a heart operation will recover is 0.3. A sample of 9 patients is selected at random.

i) Identify the probability distribution followed by the number of patients who recover.

ii) What is the probability that exactly one patient in the sample will recover.

(3 Marks)

2. a) Consider the masses of 50 oranges (measured to the nearest gram) as given in the following table.

86	95	98	107	101	87	92	93	116	105	102	105	100	105	96	111	96	117
100	106	118	101	86	103	113	101	102	104	99	99	107	97	114	88	110	100
103	108	92	109	95	101	93	118	108	113	99	106	116	101				

i) Construct a continuous frequency distribution consisting of class intervals of equal widths, with the first class being 85 - 90. (4 Marks)

ii) Draw a histogram corresponding to the distribution above together with the frequency polygon. (3 Marks)

iii) Compute the average mass of the oranges. (4 Marks)

- b) In the year 2012, the weekly household income had the following distribution

Income X (\$)	100	300	500
Probability, f(x)	0.40	0.40	0.20

Calculate

- i) The average earnings, $E(X)$ (3 Marks)
- ii) $E(3X - 2)$ (2 Marks)
- iii) The variance of X (4 Marks)

3. a) Let X be a continuous random variable with probability density function (pdf)

$$f(x) = \begin{cases} \frac{1}{2}x, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find

- (i) Mean and variance of X .

And hence

- (ii) Mean and variance of $1 - 2X$

(10 Marks)

- b) Let X be a normally distributed random variable with mean 8 and variance 16.

Find

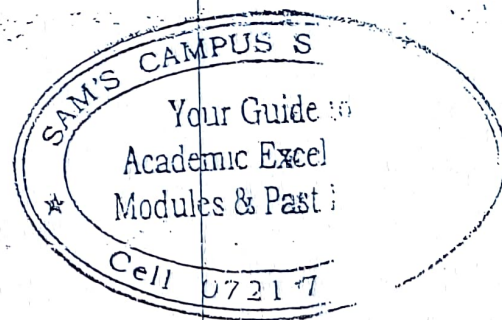
- i) $P(-5 < X < 10)$
- ii) $P(X \geq 15)$
- iii) $P(X \leq 10)$

(10 Marks)

4. a) Explain the meaning of the following terms used in estimation.

- i) Point estimate
- ii) Interval estimate

(2.2 Marks)



- b) A study claims that students in a particular college spend an average of 18 hours per week on leisure activities. The administration wanted to test this claim. A sample of 10 students selected at random gave their responses as follows (in hrs): 10, 16, 19, 22, 23, 25, 26, 33, 38 and 41

Obtain the point estimates of true population mean and true standard deviation of the time spent per week on leisure activities by all students in the college.

(6 Marks)

- c) Let X be a random variable with $P(X=x) = p_x$ such that the $\sum_{x=0}^{\infty} p_x = 1$.

- i) Define the probability generating function (pgf) of X . (2 Marks)
 ii) If X has a Bernoulli density function given by

$$P(X=x) = p_x = \begin{cases} p^x(1-p)^{1-x}, & x=0,1 \\ 0, & \text{elsewhere} \end{cases}$$

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Show that the pgf of X is given by $P(S) = 1 - p + pS$ where S is a real value. Hence find the mean and variance of X .

(8 Marks)

5. The following table shows the body weight (in kg) and plasma volume (in litres) for eight health males:

Subject	1	2	3	4	5	6	7	8
Body weight (x)	58.0	70.0	74.0	63.5	70.5	71.0	66.0	62.0
Plasma Volume(y)	2.75	2.86	3.37	2.76	2.62	3.49	3.05	3.12

- a) i) Construct a scatter diagram for these data.
 ii) What relationship does the scatter plot suggest. (4 Marks)
- b) Find the Pearson's Correlation coefficient between body weight and Plasma volume. (4 Marks)
- c) Find the regression line that would estimate the body weight given plasma volume and use it to estimate weight when the plasma volume is 2.5 litres. (6 Marks)
- d) Find the regression line that would estimate the plasma volume given the weight and use it to estimate volume when weight is 60.2kg. (6 Marks)

**FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE (ELECTRICAL AND ELECTRONICS ENGINEERING),
BACHELOR OF SCIENCE (CIVIL ENGINEERING) AND
BACHELOR OF SCIENCE (MECHANICAL ENGINEERING)**

ECU 201: ENGINEERING MATHEMATICS VI

DATE: Friday, 23rd February 2018

TIME: 11.00 a.m. - 1.00 p.m.

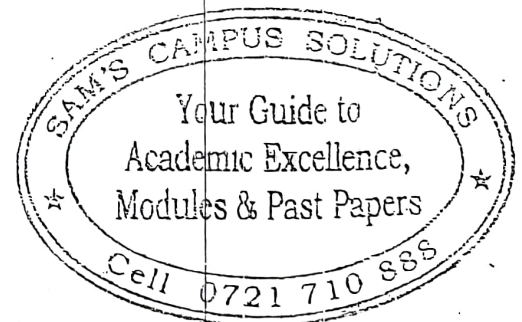
INSTRUCTIONS:

Answer Question ONE and any other two Questions.

QUESTION ONE (COMPULSORY)

- (a) The following table gives the frequency distribution of the number of personal computers sold during the past month at 40 computer stores in a big town.

Computers sold	Number of stores
4 – 12	6
13 – 21	10
22 – 30	13
31 – 39	6
40 – 48	5



Determine mean, mode, median and variance of the distribution. (10marks)

- (b) In a group of 10 students, 4 have type A personality and 6 have type B personality. If two students are selected at random from this group, what is the probability that the first of them has type A personality and the second has a type B personality? Draw a tree diagram for this experiment. (3marks)
- (c) Consider the following frequency distribution table

observation	3.2	5.8	7.9	4.5
frequency	K	k + 2	k - 3	k + 6

If the arithmetic mean is 4.876, find the value of k.

(4marks)

(d) In a production process the defective rate is 10%. Assuming a random sample of 5 items is drawn from this process, find the probability at least three of them are defective. (4marks)

(e) The number of defects on an electronic component which is used in a computerized system has been found to follow the Poisson distribution with $\lambda = 2$. Find the probability that a randomly selected electronic component will have at most two defects. (4marks)

(f) Customers buying copper parts supplied by a certain manufacturer require that the rods be between 19.9cm and 20.5 cm, inclusive. The manufacturing process is such that the actual rod lengths are well approximated by a normal distribution with mean 20.1 cm and standard deviation 0.20 cm. Determine the proportion of the manufacturer's production that is acceptable to the customer. (5marks)

QUESTION TWO

(a) The number of calls received in a particular office between 10:00 am and 11:00 am is given by the table below.

No. of telephone calls received, x	0	1	2	3	4	5	6
f(x)	0.05	0.2	0.25	0.2	0.1	0.15	0.05

- Is this a probability function? (1mark)
- Find the probability that there will be an even number of calls (2marks)
- Determine the mean and standard deviation of the number of telephone calls received per hour in that office. (7marks)

(b). The following data give the number of machines manufactured by a company for a sample of 30 days.

24	21	31	32	26	35	27	31	31	34	23	22
22	33	31	29	27	23	25	23	35	23	28	31
28	29	33	26	29	27						

- Construct a discrete frequency distribution consisting of class intervals of equal widths, with the first class being 21 - 23. (2marks)
- for what percentage of days is the number of machines produced in the interval 27 to 29? (1mark)
- Determine the mean and standard deviation of the number of machines manufactured by the company. (7marks)

QUESTION THREE

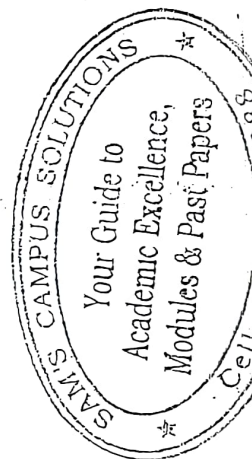
(a) Let X be a continuous random variable with probability density function (pdf)

$$f(x) = \begin{cases} \frac{5}{4}(1-x^4), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

where the random variable X denotes the clearance (in mm). the clearance is the difference between the radius of the hole drilled in a flat sheet-metal plate and a shaft inserted through the hole.

- (i) Find mean and variance of X .
 - (ii) If components with clearances larger than 0.8mm are to be scrapped, what proportion of components are scrapped? (10marks)
- (b) The diameters of ball bearings manufactured by a company are normally distributed with mean 12 mm and standard deviation 0.1 mm. Find the probability the diameters of the ball bearings will:

- (i) Exceed 12.01 mm
- (ii) Be less than 11.98 mm
- (iii) Lie between 11.98 and 12.01mm (10marks)



QUESTION FOUR

A car manufacturing company wanted to investigate how the price of one of its car models depreciates with age. The company took a sample of eight cars of this model and collected the following information on the ages (years) and prices in (hundreds of dollars) of these cars as shown below.

	258	198	180	240	171	240	222	128
Age (x)	3	6	5	2	9	6	3	8
Price (y)	86	33	36	120	19	40	74	16

- (a) Find the Pearson's correlation coefficient between ^{Age} ~~body weight~~ and ~~Plasma volume~~ ^{plasma price} (6marks)
- (b) Find the regression line that would estimate the age of a car for given price and interpret the values of the regression coefficients. Use this line predict the age when the price is \$6,400. (7marks)

- (c) Find the regression line that would estimate the price of a car for given age and interpret the values of the regression coefficients. Use this line estimate the price of a 4-year old car of this model. (7marks)

QUESTION FIVE

- (a) A random variable X has a geometric distribution whose density function is given by

$$f(x) = \begin{cases} p(1-p)^x, & x = 0, 1, 2, \dots \\ 0, & \text{elsewhere} \end{cases}$$

Show that the moment generating function of X is given by

$$M_X(t) = \frac{p}{1 - qe^t}, \text{ where } p + q = 1$$

Hence show that $\text{Var}(X) = \frac{q}{p^2}$ (12marks)

- (b) The diameters of copper shafts produced by a certain manufacturing process should have a mean diameter of 0.51 mm. The diameter is known to have a standard deviation of 0.0002 mm. A random sample of 40 shafts has an average diameter of 0.509 mm. Construct a 95% confidence interval on the mean shaft diameter. (8marks)

ECU 201

ECU 201 ENGINEERING MATHEMATICS VI CAT II

05/03/2019

Answer ALL the questions.

1. a) Given that A and B are any events such that $P(B^c) = 0.4$ and $P(A \cup B) = 0.8$, find $P(A)$ if A and B are

- i) mutually exclusive events
- ii) independent events

b) Let A and B be two events such that

$$P(A \cup B) = \frac{5}{6}, P(B^c) = \frac{1}{4} \text{ and } P(A \cap B) = \frac{1}{5}.$$

Find $P(A^c \cap B)$, $P(A^c \cap B^c)$, $P(A^c \cup B^c)$ and $P(A/B)$

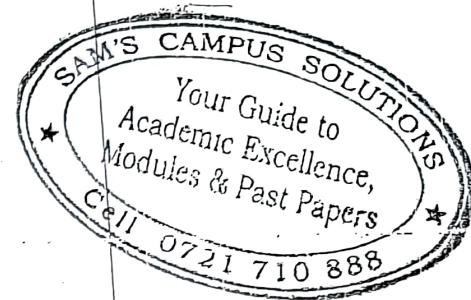
2. A continuous random variable X has probability distribution given by

$$f(x) = \begin{cases} k(2x+1), & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- a) find the value of constant k,
- b) find $P(X < \frac{1}{2})$ and $P(-1 < X < \frac{1}{2})$
- c) Determine the mean and variance of X.

3. The mean yield per plot of a crop is 17kg and standard deviation is 3kg. if the distribution of yield per plot is normal, find the percentage of plots giving yields:

- a) Between 15.5kg and 20kg
- b) More than 20 kg
- c) Less than 16kg



ANSWER ALL THE QUESTIONS

1. Let X be a continuous random variable with probability distribution given by

$$f(x) = \begin{cases} kx & \text{for } 0 < x < 8 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^8 kx \, dx = 1$$

Where k is constant.

- (i) Obtain the value of k .
(ii) Find the values of $\Pr(2 \leq X \leq 4)$ and $\Pr(-4 \leq X \leq 2)$.

2. Let X be a random variable assuming values -3, 6 and 9. If the probability distribution of X is defined by

x	-3	6	9
$f(x)$	1/6	1/2	1/3

$$\text{Var}(X) = [E(X^2) - [E(X)]^2]$$

Obtain $E(X)$, $E(X^2)$ and $\text{Var}(X)$

3. The annual incomes of a group of 10,000 persons were found to be normally distributed with mean equal to Ksh. 5,200 and standard deviation equal to Ksh. 600. Find the number of persons having income between Ksh. 4,000 and Ksh. 5,500.
4. The probability that a planted radish seed germinates is 0.90. A gardener plants ten seeds. Let X denote the number of radish seeds that successfully germinate.
- (i) What is the probability that no seed germinate? 0.1^9
- (ii) What is the probability that at least eight seeds germinate? $P(X \geq 8)$
- (iii) What is the average number of seeds the gardener could expect to germinate? 9
5. Masses of KU students have known variance of 25kg. A sample of 144 students is drawn at random, and their mean mass was found to be 64.5kg. Determine correct two decimal places the 90% confidence interval for the true mean value μ of the masses of the students.

CAT I SEM I 2018/201

Answer ALL the questions

1. The marks obtained by 50 students are given below.

41 23 56 41 40 55 48 52 40 29
40 40 56 46 32 51 54 38 69 73
54 40 29 75 54 25 57 35 22 40
86 32 34 47 25 66 49 42 31 30
52 41 29 24 33 38 27 63 32 31

- a) Construct a frequency distribution table by using the following class intervals 20-29, 30-39, 40-49, 50-59, 60-69, 70-79 and 80-89. (2 Marks)
- b) Using the table construct in (a) compute:
i) first quartile Q_1 (3 Marks)
ii) the quartile Q_3 and interquartile range (3 Marks)
iii) Standard deviation. (3 Marks)
iv) the mode (2 Marks)

2. An algebra test was given to 300 high school students of whom 100 were boys and 200 girls. The results were as follows:

Males Boys	100 = n_1	$\bar{x}_1 = 60$	$S_1 = 5.0$
Females Girls	200 = n_2	$\bar{x}_2 = 63$	$S_2 = 5.4$

- (a) Obtain the mean and the standard deviation of the combined data (3 Marks)
(b) Which of the two distributions is more variable? (3 Marks)
3. The mean monthly wage of worker in a certain factory was recorded as Kshs.5,600 and variance Kshs.1,458. But the wage of a worker was wrongly noted as Kshs.5,500 instead of Kshs.5,800. What would be the correct mean monthly wage and correct variance? (3 Marks)

Total No. of workers = 200

4. The following data refer to the ages(X years) and blood pressure(Y) of ten persons.

X	52	63	45	36	72	65	47	25	30	47
Y	62	53	51	52	25	79	43	60	64	35

- (a) Obtain Pearson correlation coefficient between ages(X) and blood pressure(Y) (4 Marks)
(b) Determine a regression line that would be use to predict the age of a person given his/her blood pressure. Use this line to estimate the age of a person whose blood pressure is 66. (4 Marks)

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2020/2021

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE BACHELOR OF SCIENCE (BIOSTATISTICS)

SST 204: PROBABILITY AND STATISTICS I

DATE: Thursday, 17th June 2021

TIME: 8.00 a.m. - 10.00 a.m.

INSTRUCTIONS: Answer Question ONE and any other TWO Questions.

Question 1 (30 Marks)

- (a) Differentiate between discrete and continuous random variables (2 Marks)
- (b) An individual who has a vehicle insurance cover from a certain company is randomly selected. Let Y be the number of moving violations for which the individual was cited during the last 3 years. The probability distribution function of Y is

Y	0	1	2	3
$P(Y=y)$	0.60	0.25	0.1	0.05

Compute the mean of Y

- (i) The mode (3 Marks)
- (ii) The median (3 Marks)
- (c) Let X be a continuous random variable with a density function $f(x)$. Give two conditions for $f(x)$ to be a probability density function of X . (2 Marks)
- (d) The time taken, to complete a given task, X is modeled by a probability density function given by $f(X=x) = \begin{cases} kx, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$ where k is a constant. $E(2x+3) = 3 + 2E(X)$ (2 Marks)
- (i) Find k , hence (2 Marks)
- (ii) determine mean $E(2X+3)$ (3 Marks)
- (iii) The cumulative density function $E(3+2X)$ (3 Marks)

- (e) A commercial bank is planning to market a new product. According to the financial analysis, it will earn an annual profit of \$ 2.5 million if this product has high sales, annual profit of \$ 0.2 million if the sales are average and it will lose \$ 0.3 million a year if the sales are low. The probabilities of these three scenarios are 0.4, 0.25 and 0.35 respectively. Let X be the profits, in millions earned per annum by the company from this product.

- (i) Write the probability distribution of X . (2 Marks)
- (ii) Calculate the standard deviation of the profits, X (3 Marks)

$$Var(2x+3) = E((2x+3) - E(2x+3))^2$$
$$E(2x+3)^2 = E(2x+3)^2$$
$$E(2x+3)^2 = E(2x+3)^2$$

(f) A company insures homes in 3 cities; Nairobi (N), Mombasa (M) and Kisumu (K). The losses occurring in these three cities are independent. The moment generating functions for the loss distributions of the cities are

$$M_M(t) = (1 - 2t)^{-2}, \quad M_N(t) = (1 - 2t)^{-3}, \quad \text{and} \quad M_K(t) = (1 - 2t)^{-4} \quad \text{respectively.}$$

Let X represent the combined losses, in dollar units, from the 3 cities. Determine

- (i) The expected value of X, $E(X)$
- (ii) The variance of X.

(3 Marks)

(4 Marks)

Question 2 (20 Marks)

(a) An audit company pays its 500 employees an average wage of \$ 15.90 an hour with a standard deviation of \$1.50. If the wages are approximately normally distributed and paid to the nearest cent, find

- (i) The percentage of the workers who receive wages between \$ 13.75 and \$ 16.22 an hour inclusive

(4 Marks)

- (ii) the number of employees who earn more than \$15.50

(3 Marks)

(b) Let the random variable X represent the number of defective parts for a machine when 3 parts are sampled from a production line tested. The following is the probability distribution of X

X	0	1	2	3
P(X)	0.51	0.38	0.10	0.01

- (i) Find the mean of defective parts

(2 Marks)

- (ii) the standard deviation

(3 Marks)

(c) The time taken to perform a particular task, t hours has the probability density function

$$f(t) = \begin{cases} 10kt^2 & , 0 \leq t \leq 0.6 \\ 9k(1-t) & , 0.6 \leq t \leq 1.0 \\ 0 & , \text{otherwise} \end{cases}$$

Where k is a constant.

Determine:

- (i) the value of k and sketch the graph of this distribution
- (ii) the expected time

(3 Marks)

(5 Marks)

Question Three (20 Marks)

(a) Obtain the moment generating function of a binomial distribution and hence find the mean and variance of a binomial distribution.

(8 Marks)

(b) In each of four matches, Gor Mahia has 60% chance of winning the match. Determine the probability that

- i. Gor Mahia will win 2 matches
- ii. Gor Mahia will win at least 1 match
- iii. Gor Mahia will win a majority of matches

(4 Marks)

(1 Mark)

(1 Mark)

(c) IQ are measured on a scale which is normally distributed with mean 100 and variance 225. The Mshindi scholarship program awards scholarships to people with the top 1% of the IQ. What would they have to set as the cut off point to allow this to happen?

(6 Marks)

Question 4 (20 Marks)

- (a) A random variable X with is defined to have a Poisson distribution if the p.d.f is given by

$$f(x) = \frac{e^{-\mu} \mu^x}{x!}, x = 0, 1, 2, \dots$$

- (i) Explain the meaning of the parameter μ (2 Marks)

- (ii) Show that $E(X) = \mu$ (5 Marks)

- (b) Assume that the average number of days schools are closed due to floods in the month of April in a county in Kenya is 4. Assuming a Poisson distribution given by

$$f(x) = \begin{cases} \frac{e^{-4} 4^x}{x!}, & x = 0, 1, 2, \dots \\ 0, & \text{otherwise} \end{cases}$$

Find the probability that the schools in this district will close for 6 days during floods.

(5 Marks)

- (c) A mathematician has established that the probability of success sale for a government treasury bond, X in the market will be 0.6. Suppose the probability distribution of the discrete

random variable X is defined by

$$f(x) = \begin{cases} 0.6^x (0.4)^{1-x}, & x = 0, 1. \\ 0, & \text{otherwise} \end{cases}$$

- (i) Show that the moment generating function of X is given by $M_x(t) = (0.4) + 0.6e^t$

(4 Marks)

- (ii) Find the mean and the variance of X

(4 Marks)

Question 5 (20 Marks)

- (a) Suppose X has a gamma distribution with parameters α and β having the probability

$$\text{function } f(x) = \begin{cases} \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} e^{-\frac{x}{\beta}}, & x > 0 \\ 0, & \text{elsewhere} \end{cases} \quad \text{where } \Gamma(\alpha) = (\alpha-1)!$$

Show that $E(X^k) = \frac{\Gamma(\alpha+k)}{\Gamma(\alpha)} \beta^k$ where k is a positive integer. Hence find the mean

and variance of X

(10 Marks)

- (b) Suppose X has a beta distribution with parameters α and β having the probability

$$\text{function } f(x) = \begin{cases} \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases} \quad \text{where } B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

Show that mean of X is $\frac{\alpha}{\alpha + \beta}$

(5 Marks)

- (c) A random variable X has probability density function

$$f(x) = \begin{cases} \frac{x}{2}, & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

If $Y = 1 - 2X$ obtain the density function of Y

(5 Marks)

(d) On the average, a certain computer part lasts ten years. If the length of time that the computer part lasts is X , then X is exponentially distributed.

Write down the density function of X and find the probability that a computer part will last between 7 and 12 years.

(5 Marks)



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2018/2019
SPECIAL/SUPPLEMENTARY EXAMINATION FOR THE
DEGREE OF BACHELOR OF SCIENCE (CIVIL, MECHANICAL, ELECTRICAL,
AEROSPACE, BIOMEDICAL AND BIOSYSTEMS ENGINEERING)
ECU 201: ENGINEERING MATHEMATICS VI

DATE: MONDAY, 28TH OCTOBER, 2019

TIME: 11.00 A.M. - 1.00 P.M.

INSTRUCTIONS:

Answer Question ONE and any other TWO Questions

$$\sum = \sqrt{\text{Var}}$$
$$s^2 = 18$$

QUESTION ONE

- (a) The following data gives the number of ball bearings manufactured by a company per day for a sample of 12 days:

23 22 47 13 15 23 19 21 23 28 25 23

Calculate mode, median and variance for these data. (5marks)

- (b) The following table gives the number of children less than 16 years of age for 28

randomly selected families:

3	1	1	1	2	0	2	0	0	2	1
2	0	1	0	0	2	1	1	3	2	2
2	2	0	1	0	2					

- (i) Prepare a frequency distribution table using single-valued classes (3marks)

- (ii) Calculate average number of children less than 16 years of age per family

(3marks)

- (iii) How many families in this sample have 2 or 3 children less than 18 years of age. (2marks)

- (c) It is known that the average number of male mates a queen bee has per minute is 2.7. Using Poisson distribution, obtain the probability that number of male mates of a bee will be exactly 3 per minute. (4marks)

- (d) In a particular factory, 56% of the employees are black, 95% are male and 53% are black males. What is the probability that a randomly selected employee will be black or male. (3marks)

- (e) In a statistics examination, the marks of students were found to be normally distributed with mean 55 and variance 225. Calculate the probability that

- (i) a student obtained less than 40 marks

- (ii) students obtained marks between 40 and 70 (4marks)

- (f) in a production process the defective rate is 10%. Assuming a random sample of 5 items is drawn from this process, find the probability at least three of them are defective. (4marks)

QUESTION TWO

- (a) The following data gives the number of machines manufactured by a company for a sample of 30 days.

22 27 30 23 28 32 27 23 27 30
 32 23 27 27 24 35 30 28 35 29
 23 26 24 34 34 30 26 24 27 30

- Construct a frequency distribution table using the classes 21 – 23, 24 – 26, 27 – 29, 30 – 32 and 33 – 35 (4marks)
- Compute the mode and median of the distribution. (6marks)
- Find the average number of manufactured machines and the standard deviation. (7marks)
- Draw the histogram corresponding to the distribution above together with the frequency curve. (3marks)

QUESTION THREE

- (a) The probability distribution of the number of times the manufacturing process is recalibrated during a week whenever the quality of the component produced falls below certain specifications is shown below:

Number of calibrations per week	0	1	2	3
Probability	0.25	0.35	0.15	0.25

- Find the mean and standard deviation of the number of calibrations
- Find the probability that the number of calibrations for this manufacturing process during a given week is
 - Exactly 2
 - More than 2
 - At most 2 (10marks)

- (a) The probability density function of random variable X is given by

$$f(x) = \begin{cases} \frac{k}{\sqrt{x}}, & 0 < x < 4 \\ 0, & \text{elsewhere} \end{cases}$$

Where k is a constant.

- Show that $k = 1/4$.

Find

FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE
IN CIVIL ENGINEERING

ECU 201: ENGINEERING MATHEMATICS VI

DATE: TUESDAY 6TH DECEMBER 2016

TIME: 8.00 A.M. - 10.00 A.M.

INSTRUCTIONS: Answer Question ONE and any other two Questions.

Q1. (a) The following data gives the number of vehicles owned by each of the 30 families in

a city: 2, -3, 1, 1, 1, 2, 4, -3, 1, 1, -3, 2, 2, 1, 2, -3, 2, -3, 1, 3, 4, 1, 3, 4, 1, 1, 1, 1, -3, 2, 2

(i) Prepare a frequency distribution table for this data set using single-valued classes

(ii) Calculate the mean number of vehicles per family

(iii) Find the percentage of the families in this sample having two or three vehicles

(3,3,2 marks)

(b) The following are the ages of 15 employees of a manufacturing company:

65, 38, 52, 27, 24, 45, 49, 50, 26, 37, 44, 52, 41, 60, 39

Calculate the mode, median and standard deviation (5marks)

(c) In a group of 10 adults, 4 have type A personality and 6 have type B personality. If two

persons are selected at random from this group of 10, what is the probability that the

first of them has type A personality and the second has type B personality? Draw a

tree diagram for this experiment. (4marks)

(d) The probability that a family owns a desktop computer is 0.78, that it owns a CD player

is 0.70 and that it owns both the computer and a CD is 0.60. Find the probability that a

randomly selected family owns a computer or a CD player. (3marks)

(e) A student's campus newspaper in a certain university contains an average of 1.2

typographical errors per page. Find the probability that a randomly selected page of this

newspaper will contain exactly 4 typographical errors using the Poisson distribution.

(4marks)

(f) The comprehensive strength of samples of cement can be modeled by a normal

distribution with a mean of 7000 kg/cm^2 and a standard deviation of 100 kg/cm^2 .

(i) Find the probability of a sample's strength that is less than 7250 kg/cm^2

(ii) Find the probability that a sample's strength is between 6800 and 6900 kg/cm^2

(6marks)

QUESTION TWO

(a) The following data gives the number of car oil changes made at a service station for a sample of 30 days.

23 31 26 24 34 34 30 26 24 27
21 27 30 23 28 32 27 23 27 30
33 23 27 27 24 35 30 28 35 29

- Construct a frequency distribution table using the classes 21 – 23, 24 – 26, 27 – 29, 30 – 32 and 33 – 35. (4marks)
- Compute the mode and median of the distribution. (6marks)
- Find the average number of oil changes and the standard deviation. (7marks)
- Draw the histogram corresponding to the distribution above together with the frequency polygon. (3marks)

QUESTION THREE

(a) The following table shows the probability distribution of the number of breakdowns per week for a machine.

Breakdowns per week	0	1	2	3
Probability	0.15	0.20	0.35	0.30

- Find the mean and standard deviation of the number of breakdowns.
- Find the probability that the number of breakdowns for this machine during a given week is
 - Exactly 2
 - More than 1
 - At most 1

(10marks)

(b) The thickness of a conductive coating in micrometers has a density function(pdf)

$$f(x) = \begin{cases} \frac{600}{x^2}, & 100 \mu\text{m} < x < 120 \mu\text{m} \\ 0, & \text{elsewhere} \end{cases}$$

- Find the mean and variance of the coating thickness.
- If the cost of coating is \$0.25 per micrometer of thickness on each component, find the average cost of coating per component.
- Find the probability that the coating is more than 115 micrometers.

(10marks)



QUESTION THREE (20 MARKS)

The following table shows the ages in years (X) and blood pressure (Y) of 10 persons.

X:	56	42	36	47	49	42	60	72	63	55
Y:	147	125	118	128	145	140	155	164	149	150

- Find the correlation coefficient between age and blood pressure and comment on your result. (8 Marks).
- Determine the least square regression equation of age on blood pressure. (5 marks).
- Estimate the age of a person whose blood pressure is 135. (2 Marks)
- Obtain the Spearman's rank correlation coefficient. (5 Marks)

QUESTION FOUR (20 MARKS)

- The probability density function of a continuous random variable X is given by

Where k is a constant

- Show that $k = \frac{1}{6}$ (2 Marks).
 - Determine the mean and variance of X. (6 Marks)
 - Find $P(X < 5)$ (2 Marks).
- As a result of tests on 20,000 electric bulbs manufactured by a company it was found out that the lifetime of the bulb was normally distributed with an average life of 2,040 hours and standard deviation of 60 hours. On the basis of the information estimate the number of the bulbs that are expected to burn for:
 - More than 2,150 hours (3 Marks).
 - Less than 1,960 hours (3 Marks)
 - Between 1,960 hours and 2,150 hours. (4 Marks)

QUESTION FIVE (20 MARKS)

- The following readings give the weights of 6 students (in kgs) selected at a random from university students in a certain university:

50, 52, 55, 60, 65, 64

- Find the estimates of the true population mean weight μ and true standard deviation of all the students. (6 Marks)
- Construct a 95% confidence interval for the true population mean weight μ . (4 Marks).

b) A random variable Y has a poisson distribution whose probability distribution function is given by;

$$f(y) = \begin{cases} \frac{e^{-\lambda} \lambda^y}{y!} & , y = 0, 1, 2, \dots, \lambda > 0 \\ 0 & \text{otherwise} \end{cases}$$

Show that the moment generating function of Y is given by

$$M_Y(t) = e^{-\lambda(1-e^t)}$$

Hence find the mean and variance of Y

(10 Marks.)

ANSWER ALL THE QUESTIONS

15-19	111	3
20-24	111	3
25-29	11	
30-34	1111	4
35-39	111	3
40-44	111	3
45-49	111	3
50-54	1	1
55-59	1	1

1. The following are marks scored by 30 students in a statistics cat

~~23~~ 47 ~~51~~ ~~37~~ ~~21~~ ~~21~~ 30 ~~31~~ ~~39~~ 41
~~30~~ ~~38~~ ~~26~~ ~~22~~ 44 56 ~~18~~ ~~30~~ ~~31~~ ~~37~~
~~37~~ 48 ~~17~~ ~~16~~ ~~31~~ 41 ~~48~~ ~~28~~ ~~32~~ ~~31~~

a) Construct a frequency distribution table by using the following class intervals

15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59

b) Compute the mean, mode and median of the group data in (a),

(6 marks)

c) Calculate the standard deviation, first quartile, third quartile and interquartile range of the group data in (a)

(6 marks)

2. The age and blood pressure of 10 women are

Age (x yrs)	95	40	34	45	47	41	61	70	61	53
Blood pressure (y)	145	123	116	126	143	141	153	162	147	151

a) Find the correlation coefficient between age and pressure

(4 marks)

b) Obtain the least square regression line equation of blood pressure on age.

(4 marks)

c) Estimate the blood pressure of a woman whose age is 46 years.

(2 marks)

3. A CAT was given to 300 students of whom 100 were boys and 200 girls. The results were as follows:

Boys	100	$\bar{x}_1 = 72$	$S_1 = 6.3$
Girls	200	$\bar{x}_2 = 74$	$S_2 = 6.5$

a) Obtain the mean and the standard deviation of the combined data.

(4 marks)

b) Which of two distribution is more variable?

(2 marks)